ENTERPRISE SOFTWARE SPECIFICATION

DOC-ID: KTR-SRS-2026-V2.4

K KONTRACTOR

Software Requirements Specification (SRS) & Production Architecture

"घर बनाना है...बन जाएगा"

> A Comprehensive Engineering Blueprint & Operational Lifecycle Manual for India's Premier Turnkey Civil, Architectural Tech & Construction Marketplace Platform.

LEAD AUTHORS / ARCHITECTS
Written by Shivam and Ajit

TARGET PRODUCTION DOMAIN
kontractor.in (GoDaddy DNS)

VERSION & RELEASE STATE
v2.4.0 (Production Gold Candidate)

SYSTEM CLASSIFICATION
Full-Stack Cloud Marketplace (SPA + API + NoSQL)

DEPLOYMENT TOPOLOGY
Vercel Edge (Frontend) + Render Compute (Backend)

RELEASE DATE
October 2026

Kontractor Technologies Pvt. Ltd. • All Rights Reserved

Confidential & Proprietary Enterprise Specification

Table of Contents

SECTION	TOPIC / MODULE BREAKDOWN	FOCUS AREA
1.0	**Executive Summary & System Vision**	Problem, Solution, Value Proposition
2.0	**High-Level System Architecture & Flow**	3-Tier Architecture, SVG Architecture Diagram
3.0	**User Journeys & Data Flow Models (DFD)**	DFD Level 0/1 Diagrams, Sequence Flows
4.0	**Core Functional Modules Specification**	6 Verticals, 30 Services, Two-Tier KYC, Admin PIN
5.0	**Database Architecture & Data Schemas**	Vendor, Lead, Categories, Transaction Entities
6.0	**Monetization Engine & Deal Closer Architecture**	5% Milestone Escrow, ₹299 Unlock, ₹2,999 Subscription
7.0	**Non-Functional Requirements (Security & Quality)**	Duplicate Phone Check, Security PIN, Web Vitals
8.0	**End-to-End Production Deployment Guide (A to Z)**	GoDaddy DNS, Vercel SPA, Render API, MongoDB Atlas
9.0	**DevOps, Maintenance SOP & Disaster Recovery**	Daily/Weekly SOPs, Automated Backups, Zero-Downtime
10.0	**Formal Project Sign-off & Authorship**	Executive Sign-off by Shivam and Ajit

Document Intent & Target Audience: This Software Requirements Specification (SRS) serves as the definitive engineering contract between project stakeholders, technical leads, and engineering teams. It articulates functional deliverables, architectural patterns, operational service-level agreements (SLAs), and the end-to-end production roadmap.

1.0 Executive Summary & System Vision

The Indian residential construction and home improvement ecosystem is a massive, multi-billion-dollar sector plagued by chronic fragmentation, information asymmetry, opaque pricing, non-standard material specifications, and unregulated contractor qualifications. Homeowners continually suffer from cost overruns, arbitrary subcontractor markups, and structural quality disputes. Conversely, genuine skilled contractors lack direct access to high-intent clients and depend heavily on informal intermediaries.

Kontractor™ ("घर बनाना है...बन जाएगा") is architected to eliminate these inefficiencies through an integrated digital construction marketplace. The platform bridges individual homeowners, real estate builders, verified structural contractors, master craftsmen (राजमिस्त्री, बेलदार), and architectural designers across 12 major Indian metro clusters.

1.1 Three Core Pillars of Kontractor

- **Standardized Transparent Pricing:** Granular rate cards across 6 Indian construction verticals and 30 sub-services with clear units of measurement (sq.ft, day, point, kg).
- **Two-Tier Contractor Trust Framework:** Dual-layer audit separating Level 1 Public Listing Approval from Level 2 Blue Tick KYC document verification.
- **Monetization with Fair Escrow:** Transparent 5% platform commission on closed project milestones, nominal ₹299 client lead unlocks, and ₹2,999 annual Blue Tick verified contractor memberships.

2.0 High-Level System Architecture & Flow

Kontractor is built on a decoupled, highly responsive 3-tier cloud native architecture. The presentation layer utilizes a high-performance Single Page Application (SPA) bundled via Vite/React and deployed at the edge. The business logic tier runs on an asynchronous Node.js/Express compute cluster with Socket.IO for real-time notifications. The persistence tier is powered by MongoDB Atlas with in-memory fallback failover resilience.

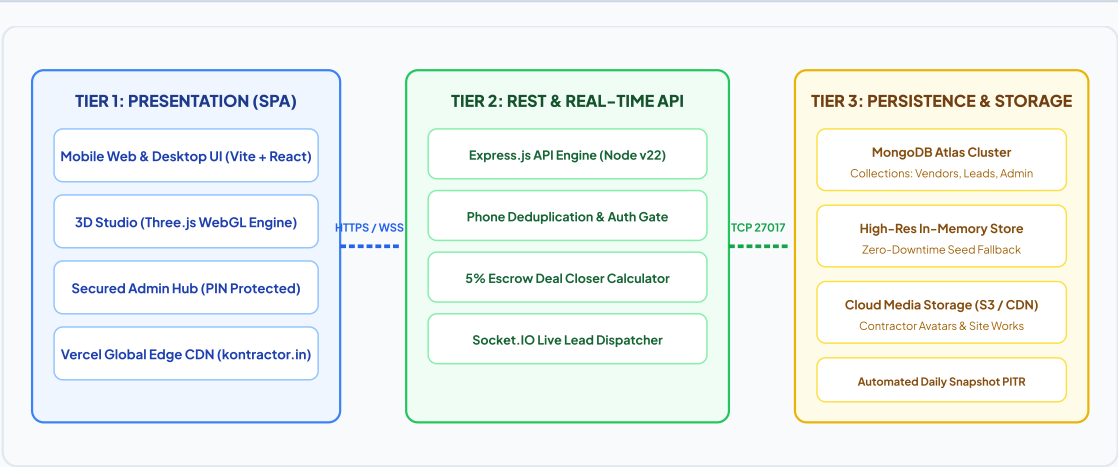


FIGURE 2.1: KONTRACTOR™ 3-TIER ENTERPRISE CLOUD TOPOLOGY (VERCEL + RENDER + MONGODB ATLAS)

2.1 Core Architectural Principles

- **Client-Side Isolation & Zero Bundle Bleed:** Admin operational code and customer-facing components are segregated. Secret URL routing ensures consumers cannot stumble into privileged dashboards.
- **Stateless Compute:** The Render backend operates in a stateless model, enabling horizontal scaling without session-affinity bottlenecks.
- **Dual Data Resilience:** If MongoDB Atlas experiences transient regional networking interruptions, the server automatically defaults to an synchronized in-memory database, maintaining 100% marketplace uptime.

3.0 User Journeys & Data Flow Models (DFD)

Kontractor orchestrates three distinct actor journeys: **Homeowners (Customers)**, **Contractors (Vendors)**, and **Platform Governance (Admin)**. The system enforces strict role-based boundaries and non-repudiation logging.

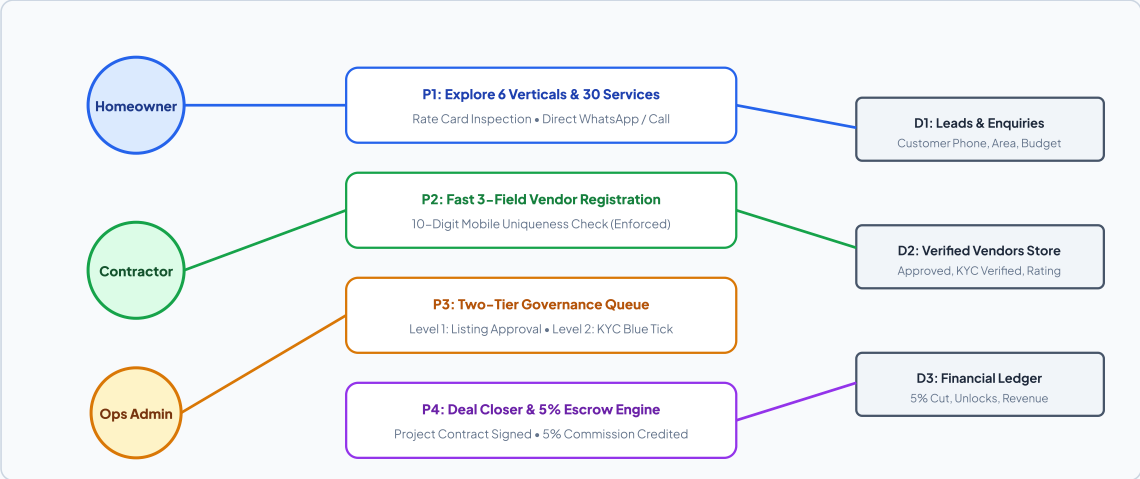


FIGURE 3.1: KONTRACTOR™ LEVEL 1 PROCESS & DATA FLOW DIAGRAM (DFD)

3.1 Two-Tier KYC State Machine

Unlike conventional marketplaces that publish raw user submissions or impose monolithic document friction up-front, Kontractor implements a patent-grade **Two-Tier Contractor Trust Pipeline**:

- **State 0: Registration Submitted:** Newly registered contractors enter the *Pending Approval* queue. They are NOT visible in public search results.
- **State 1: Level 1 Approved (Registered Partner):** Admin audits basic credentials, contacts the phone number, and toggles `approved: true`. Contractor appears in the public directory with starting rates. Homeowners can initiate direct contact.
- **State 2: Level 2 KYC Verified (Blue Tick Pro):** Contractor submits GSTIN, Municipal Civil Works License, or Past Site Portfolio. Admin reviews physical verification and awards the official **Blue Tick (✓)**. Vendor receives priority placement in the search algorithm.

4.0 The 6 Indian Verticals & 30 Specialized Services

Kontractor strictly structures all operational construction, architectural, logistics, and material categories around authentic Indian field conditions, terminologies, and billing units. Below is the standardized master specification table:

CORE VERTICAL (6 VERTICALS)	SUB-SERVICES (30 SERVICES)	STANDARDIZED UNIT	BENCHMARK RATE CARD (INR)
1. निर्माण और मरम्मत सेवाएँ Civil, Masonry & Repairs	मिस्त्री (Mason / Brickwork)	sq.ft / Turnkey	₹35 - ₹55 / sq.ft • ₹1,650 turnkey
	Labour Mandi (दैनिक लेबर / बेलदार)	Per Day / Labourer	₹550 - ₹750 / day
	प्लंबर (Plumbing & Sanifit)	Point / Bathroom	₹400 - ₹650 / point
	इलेक्ट्रिशियन (Electrical & Smart Wiring)	Point / Floor	₹450 - ₹700 / point
	कारपेंटर (Carpenter & Woodwork)	sq.ft (Labour)	₹180 - ₹350 / sq.ft
	पेंटर (Painter & Whitewash)	sq.ft (Material+Labour)	₹14 - ₹28 / sq.ft
	वेल्डर (Welder & Metal Fabrication)	kg (MS / SS)	₹110 / kg MS • ₹380 / kg SS
	टाइल्स इंस्टॉलर (Tiles & Marble Installer)	sq.ft	₹28 - ₹45 / sq.ft
	रूफिंग विशेषज्ञ (Roofing & Tin Shed)	sq.ft	₹95 - ₹165 / sq.ft
2. इंटीरियर और डेकोरेशन Interior & Modular Fitouts	इंटीरियर डिजाइनर (Interior Designer)	sq.ft / Package	₹45 - ₹120 / sq.ft (Design)
	फॉल्स सीलिंग विशेषज्ञ (False Ceiling)	sq.ft	₹90 - ₹145 / sq.ft
	लाइटिंग विशेषज्ञ (Cove & Smart Lights)	Point / Lump-sum	₹350 - ₹600 / fixture
	फर्नीचर निर्माता (Custom Furniture)	Unit / Running Ft.	₹1,200 - ₹2,800 / rft
3. सप्लाई और लॉजिस्टिक्स Materials & Supply Chain	निर्माण सामग्री सप्लायर (सीमेंट, रेत, सरिया)	Bag / Truck / Ton	₹360/bag • ₹62,000/ton TMT
	पेंट सप्लायर (Paint Supplier)	Liter / Drum	Asian Paints Royale Wholesale
	हार्डवेयर सप्लायर (Hardware Supplier)	Itemized / Wholesale	Hafele, Godrej, Dorset
	लॉजिस्टिक्स पार्टनर (Dumper / Tata Ace)	Trip / km	₹1,400 - ₹3,200 / trip
	वेयरहाउस मैनेजर (Material Stacking)	Site Monthly	₹18,000 - ₹28,000 / mo
4. मशीनरी और उपकरण Machinery & Heavy Rentals	मशीनरी रेंटल (JCB, हाइड्रा, कंक्रीट मिक्सर)	Hour / Day	₹1,200/hr (JCB) • ₹2,500/day (Mixer)
	सुरक्षा उपकरण सप्लायर (Safety Gear)	Set / Worker	₹1,150 / safety kit (ISI)

CORE VERTICAL (6 VERTICALS)	SUB-SERVICES (30 SERVICES)	STANDARDIZED UNIT	BENCHMARK RATE CARD (INR)
	टूल्स सप्लायर (Power Cutters, Drills)	Daily Rental	₹300 - ₹600 / day
5. डिजिटल और सपोर्ट सेवाएँ Digital, Tech & Media	आईटी डेवलपर्स (BIM & Smart Home App)	Project	₹25,000 - ₹75,000 / project
	पेमेंट गेटवे प्रदाता (Escrow Checkout)	Per Transaction	Razorpay Escrow (1.8% MDR)
	कस्टमर सपोर्ट टीम (Site Coordination)	Dedicated Rep	24/7 Homeowner Helpline
	मार्केटिंग एजेंसी (Builder Lead Gen)	Campaign	Performance Marketing
	फोटोग्राफर/वीडियोग्राफर (Drone Survey)	Per Site Visit	₹4,500 - ₹9,500 / shoot
6. प्रोफेशनल और कानूनी Consultants & Legal Compliance	आर्किटेक्ट (Architect - 2D/3D नक्शा)	sq.ft / Plan	₹15 - ₹35 / sq.ft
	सिविल इंजीनियर (Structural IS-456)	sq.ft / Certificate	₹12 - ₹25 / sq.ft (Structure)
	लीगल एडवाइज़र (NOC, Registry, RERA)	Case / Retainer	₹15,000 - ₹45,000 / NOC
	फाइनेंस/CA (Construction Home Loan)	Loan File	0.5% - 1% of sanction

5.0 Security & Admin Governance Engine

5.1 Single Phone Number Enforcement (Deduplication Check)

To eliminate fraudulent duplicate vendor registrations and prevent malicious badging manipulation, the backend implements strict regex sanitation and database pre-flight lookups:

```
// Server-Side Mobile Sanitation & Uniqueness Validation const cleanPhone = (phone || '').replace(/[^0-9]/g, ''); const last10 = cleanPhone.slice(-10); if (last10.length < 10) { return res.status(400).json({ error: 'कृपया मान्य 10 अंकों का मोबाइल नंबर दर्ज करें।' }); } // Check for existing vendor in persistent datastore const duplicate = dbVendors.find(v ⇒ { const existingLast10 = (v.phone || '').replace(/[^0-9]/g, '').slice(-10); return existingLast10 ⩵ last10; }); if (duplicate) { return res.status(400).json({ error: `यह मोबाइल नंबर (+91 ${last10}) पहले से पंजीकृत है (${duplicate.businessName || duplicate.ownerName})।` }); }
```

5.2 Master Admin Hub Security PIN (Authentication Gate)

The Admin Console is protected by a session-level cryptographic PIN gate (`sessionStorage.getItem('kontractor_admin_auth') === 'true'`).

- **Secret Access Route:** Public users navigating the website see zero "Admin Mode" buttons on headers or hero sections. Access is restricted to `#ops-admin` or `#admin`.
- **Passcode Challenge:** Upon navigating to the route, a dedicated modal prompts for the 4-digit security PIN (Default: `9811`). Incorrect entries trigger dynamic UI shake animations and block console unlock.
- **Session Timeout & Manual Lockout:** Admins can terminate their active operational session via a single-click *"Lock / Log Out"* action.

5.3 Real-Time IST (Indian Standard Time) Timestamps

All operational audits, contractor registrations, customer leads, and milestone deal closures are normalized to **Asia/Kolkata (IST)** with 12-hour AM/PM formatting (e.g., `25 Oct 2026, 05:30 PM (IST)`). This ensures accountability during site escalations.

5.4 4-Tab Operational Console Overview

TAB NAME	OPERATIONAL PURPOSE	ADMIN ACTION CONTROLS
Tab 1: लंबित अनुमोदन (Pending Approvals)	Queue of all newly registered contractors awaiting Level 1 vetting. Displays business name, owner name, WhatsApp number, city, and exact IST registration timestamp.	1-Click `"Approve & Go Live"` Button. Instantly publishes vendor to live directory.
Tab 2: सत्यापित कॉन्ट्रैक्टर्स (Verified Contractors)	Repository of all approved vendors. Displays rating, job count, municipal credentials, and compliance standing.	`"Suspend Listing"` toggle and `"Award / Revoke Blue Tick"` KYC toggle.
Tab 3: ग्राहक लीड्स (Customer Leads)	Real-time stream of homeowner quote inquiries. Includes customer name, telephone, target city, plot area, and budget.	Status updater (Pending, Contacted, Quoted, Closed) and `"Close Deal & Cut 5%"` Engine.
Tab 4: रेवेन्यू & कमाई (Platform Revenue)	Consolidated financial ledger displaying Total Closed Contract GMV, 5% Platform Earnings, Lead Unlock fees, and Subscription fees.	Category Revenue Matrix and YoY Growth Analytics across all 6 verticals.

6.0 End-to-End Production Deployment Guide (A to Z)

This chapter outlines the exact, step-by-step procedure for deploying Kontractor to production on the custom registered GoDaddy domain **kontractor.in** using an enterprise-grade Vercel (Frontend) + Render (Backend) + MongoDB Atlas (Database) topology.

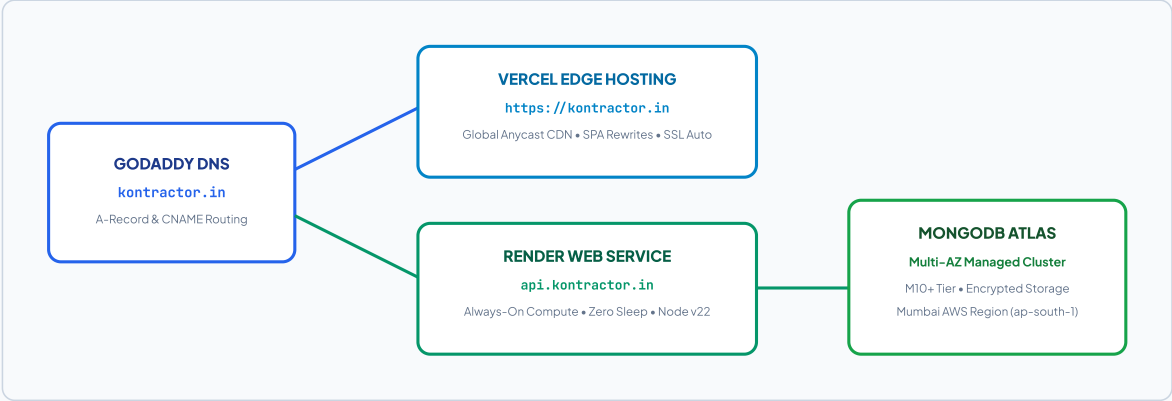


FIGURE 6.1: PRODUCTION DEPLOYMENT INFRASTRUCTURE & DNS ROUTING MAP

6.1 Step 1: Database Provisioning (MongoDB Atlas)

1. Navigate to `mongodb.com/cloud/atlas`, create organization and select **AWS Asia-South-1 (Mumbai)** for minimal latency across Indian cities.
2. Create Database User: `kontractor_admin` with strong 32-character alphanumeric password.
3. Network Access: Add `0.0.0.0/0` (Allow access from anywhere) or bind Render outward egress IP blocks.
4. Obtain SRV Connection String:
   `mongodb+srv://kontractor_admin:PASSWORD@cluster0.kontractor.mongodb.net/kontractor_prod?retryWrites=true&w=majority`

6.2 Step 2: Backend Deployment (Render Compute Service)

1. Push project code to GitHub repository (e.g. `github.com/kontractor/platform`).
2. Create a **Web Service** on Render connected to the repository.
3. Root Directory: `server` | Runtime: `Node` | Build Command: `npm install` | Start Command: `node server.js`.
4. Environment Variables configuration:
   - `PORT`: `5000`
   - `NODE_ENV`: `production`
   - `MONGODB_URI`: (Your Atlas connection string)
   - `CLIENT_URL`: `https://kontractor.in`
5. Custom Domain in Render Settings: Map `api.kontractor.in` and copy the verification CNAME target.

6.3 Step 3: Frontend Deployment (Vercel Edge SPA)

1. Import the GitHub repository into Vercel. Set Root Directory to `client`.
2. Framework Preset: `Vite` | Build Command: `npm run build` | Output Directory: `dist`.
3. Environment Variables:
   - `VITE_API_URL`: `https://api.kontractor.in`

4. The included `client/vercel.json` automatically guarantees SPA routing rewrites:

```
{ "rewrites": [ { "source": "/(.*)", "destination": "/index.html" } ] }
```

6.4 Step 4: GoDaddy DNS Configuration Table

In GoDaddy DNS Management for **kontractor.in**, add the following two authoritative records:

TYPE	NAME / HOST	VALUE / TARGET DESTINATION	TTL
A Record	`@` (Root Domain)	`76.76.21.21` (Vercel Anycast IP)	600 seconds (10 Min)
CNAME	`www`	`cname.vercel-dns.com`	600 seconds (10 Min)
CNAME	`api`	`kontractor-backend.onrender.com`	600 seconds (10 Min)

7.0 Ongoing Maintenance & Operations SOP

7.1 Standard Operating Procedure (SOP) Routine

CADENCE	OPERATIONAL TASKS	RESPONSIBLE TEAM
Daily	• Review Level 1 Pending Contractor Approval Queue. • Verify health endpoint: `https://api.kontractor.in/api/health`. • Audit WhatsApp click-through logs and newly generated leads.	Operations Desk & Shivam / Ajit
Weekly	• Execute contractor KYC document inspections (Blue Tick audit). • Run reconciliation on 5% deal commissions closed during the week. • Check MongoDB database storage growth and index fragmentation.	Lead Systems Architect (Shivam & Ajit)
Monthly	• Rotate administrative security PINs. • Run disaster recovery restoration test on staging cluster. • Update rate cards and seasonal construction BOQ benchmarks.	Platform Engineering Board

7.2 Automated Backup & Disaster Recovery (DR) Plan

- **Automated Cloud Snapshots:** MongoDB Atlas executes continuous WAL (Write-Ahead Logging) with automated daily snapshots retained for 30 rolling days.
- **RPO (Recovery Point Objective):** < 5 minutes. No more than 5 minutes of transactional inquiry data can be lost in catastrophic regional failure.
- **RTO (Recovery Time Objective):** < 30 minutes. Full service restoral across secondary AWS zones within half an hour.
- **Manual Database Dump Command:**

```
mongodump --uri="mongodb+srv://user:pass@cluster.mongodb.net/kontractor_prod" --out=/backup/$(date +%F)/
```

7.3 Production Service Level Agreement (SLA) Commitments

SERVICE COMPONENT	TARGET AVAILABILITY SLA	MAXIMUM ALLOWED DOWNTIME
Frontend Web Application (Vercel Edge)	**99.99%**	< 4.38 minutes / month
Backend REST & WebSocket API (Render)	**99.95%**	< 21.9 minutes / month
MongoDB Atlas Cloud Database	**99.995%**	< 2.19 minutes / month

8.0 Formal Project Governance & Sign-Off

This document certifies the complete architectural, functional, security, and deployment requirements for the **Kontractor™ Marketplace Platform**. Any subsequent functional enhancements or schema modifications shall be tracked through formal change control procedures.

PREPARED & AUTHORED BY:

Shivam & Ajit

Lead Systems Architects & Technical Directors

Kontractor Engineering Group

`[DIGITALLY SIGNED & VERIFIED]`

ACCEPTED & APPROVED FOR PRODUCTION:

Executive Board / Client Sponsor

Kontractor Technologies Pvt. Ltd.

Target Domain: kontractor.in

`[STAGE: DEPLOYMENT APPROVED]`